

AI Powered Debt Relief & Financial Recovery Platform

Project Description:

The AI Powered Debt Relief & Financial Recovery Platform is an intelligent web application designed to assist individuals facing financial challenges by providing personalized debt management and recovery solutions. The platform enables users to securely manage their loan information, analyze their financial condition, and receive AI-powered recommendations for debt settlement and financial planning.

The application integrates the Antigravity AI Platform to generate personalized settlement strategies and professional negotiation letters based on the user's financial profile. The system also includes user authentication, loan management, financial analysis, AI-driven recommendations, and history tracking to provide a complete financial assistance solution.

Built using React.js for the frontend, FastAPI for the backend, SQLite with SQLAlchemy for database management, and powered by the Antigravity AI Platform, the application delivers a responsive, secure, and user-friendly experience while promoting responsible financial recovery.

Scenarios

Scenario 1: Debt Settlement Recommendation

A user logs into the platform and enters details such as monthly income, outstanding loan amount, EMI, and overdue period. The AI analyzes the financial information and recommends a suitable debt settlement percentage, repayment strategy, and financial recovery plan based on the user's current financial condition.

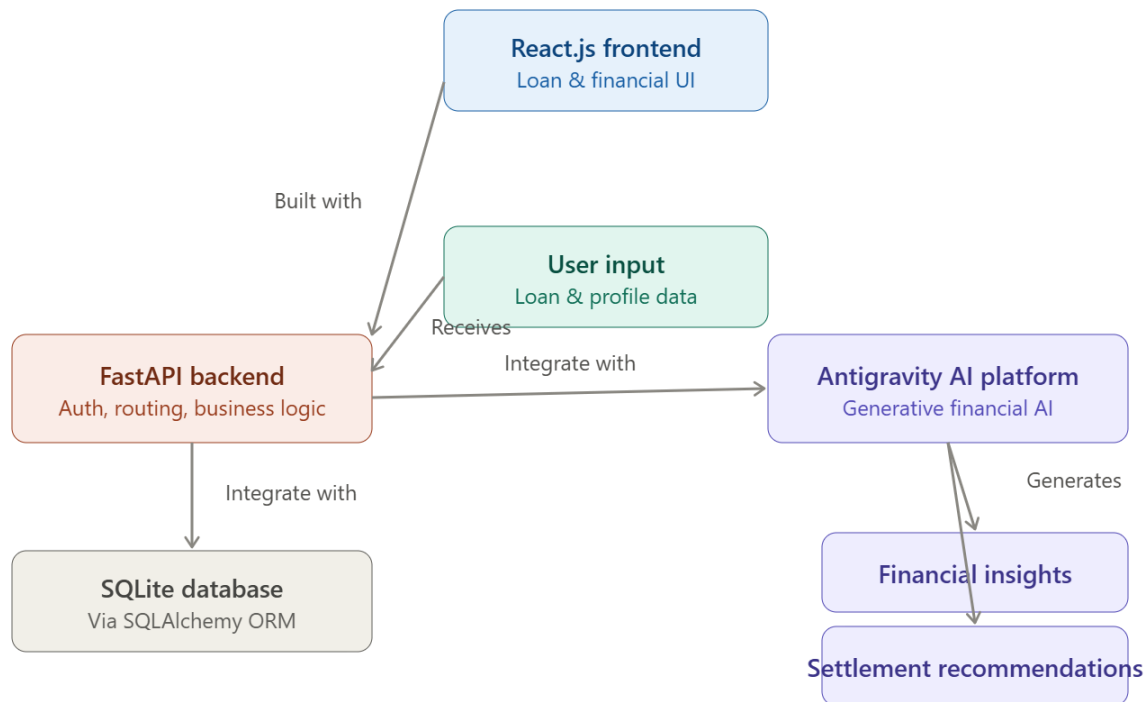
Scenario 2: AI Negotiation Letter Generation

A borrower facing financial hardship wants to request a loan settlement from the bank. After submitting the required financial details, the AI automatically generates a professional debt negotiation letter that clearly explains the user's financial situation and requests a fair settlement plan.

Scenario 3: Financial Health Monitoring

A returning user accesses the dashboard to review previously submitted loan information and AI-generated recommendations. The platform displays financial analysis, settlement history, and personalized suggestions that help the user monitor progress and make informed financial decisions.

Technical Architecture:



Description: The **AI Powered Debt Relief & Financial Recovery Platform** follows a **three-tier architecture** consisting of the **Frontend**, **Backend**, and **Database** integrated with the **Antigravity AI Platform**. The **React.js** frontend provides an interactive user interface for loan management and financial analysis. The **FastAPI** backend processes user requests, handles authentication, business logic, and communicates with both the database and the AI platform. **SQLite** with **SQLAlchemy** stores user, loan, and settlement data securely. The **Antigravity AI Platform** generates financial insights, settlement recommendations, and negotiation letters, delivering intelligent and personalized debt management solutions.

ER Diagram:

The **Entity Relationship (ER) Diagram** represents the database structure of the **AI Powered Debt Relief & Financial Recovery Platform**. It consists of five main entities: **Users**, **Financial Profiles**, **Loans**, **AI History**, and **Settlement Records**.

- **Users** stores user registration and login details.
- **Financial Profiles** contains users' income, expenses, debts, and financial health score.
- **Loans** stores loan-related information such as loan amount, outstanding amount, interest rate, and due date.
- **AI History** maintains AI-generated negotiation strategies, settlement letters, and responses.
- **Settlement Records** stores AI-predicted settlement recommendations and suggested repayment amounts.

The relationships between these entities enable efficient management of user data, financial analysis, loan tracking, AI-generated recommendations, and settlement history, ensuring organized and secure data storage within the system.

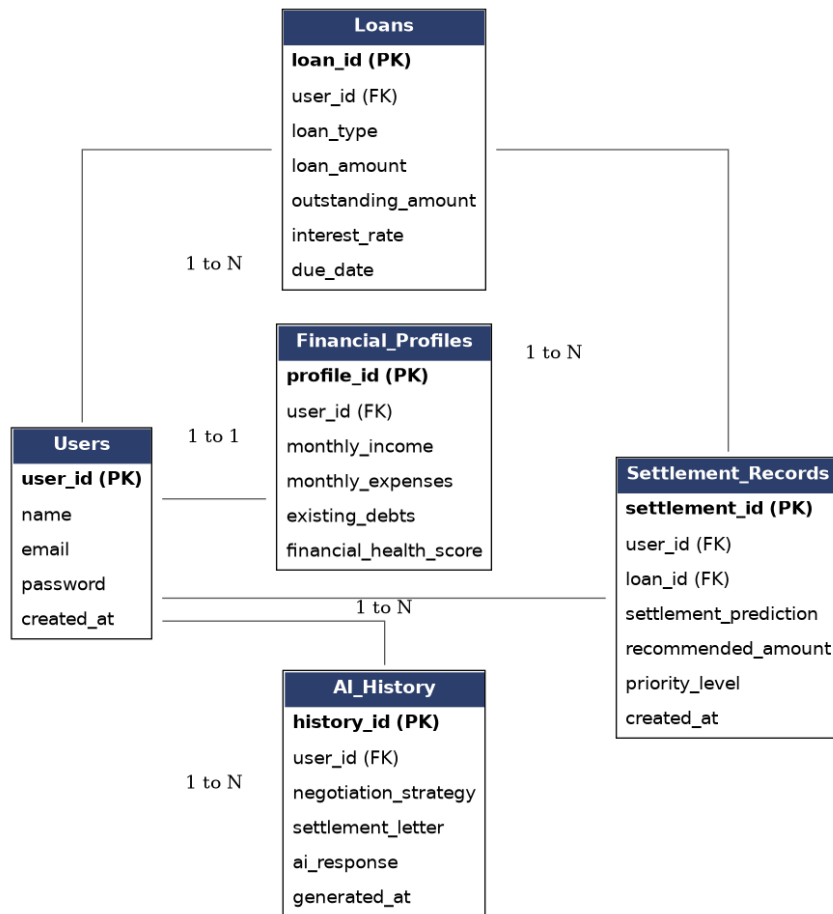


Figure: FinRelief AI entity-relationship diagram

Pre-requisites:

- **Flask Framework Knowledge:** [Flask Documentation](#)
- **Groq API Familiarity:** [Groq API Documentation](#)
- **HTML, CSS, and JavaScript Skills:** [W3Schools HTML/CSS/JavaScript Tutorials](#)
- **Python Programming Proficiency:** [Python Documentation](#)
- **Version Control with Git:** [Git Documentation](#)
- **Development Environment Setup:** [Flask Installation Guide](#)
- **Ngrok for Public Deployment:** [Ngrok Documentation](#)

Project Workflow:

Milestone 1: AI Platform Configuration and System Architecture

- **Activity 1.1:** Configure the Antigravity AI Platform and securely integrate the API credentials into the project environment.
- **Activity 1.2:** Research and select the appropriate AI capabilities for financial analysis, debt settlement recommendations, and negotiation letter generation.
- **Activity 1.3:** Define the system architecture, illustrating the interaction between the React frontend, FastAPI backend, SQLite database, and Antigravity AI Platform.
- **Activity 1.4:** Set up the complete development environment by installing the required software, libraries, and project dependencies.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the core modules including User Authentication, Loan Management, Financial Analysis, AI Settlement Recommendation, and Negotiation Letter Generation.
- **Activity 2.2:** Implement the FastAPI backend to process user requests, validate financial data, communicate with the database, and integrate AI services.

Milestone 3: Backend API Development

- **Activity 3.1:** Develop REST APIs for authentication, loan management, financial analysis, AI recommendation generation, and history management.

Milestone 4: Frontend Development

- **Activity 4.1:** Design and develop responsive user interfaces using React.js and Vite.
- **Activity 4.2:** Connect frontend components with backend APIs and dynamically display financial analysis and AI-generated recommendations.

Milestone 5: Deployment

- **Activity 5.1:** Prepare the application for deployment by configuring the production environment and installing all required dependencies.
- **Activity 5.2:** Deploy the application to a cloud hosting platform and verify all application features.

Milestone 6: Conclusion

Milestone 1: AI Platform Configuration and System Architecture

In this milestone, the primary objective is to configure the Antigravity AI Platform and establish the overall architecture of the application. This includes integrating AI services, defining communication between the frontend, backend, and database, and preparing the development environment for implementing intelligent financial analysis features.

Activity 1.1: Configure the Antigravity AI Platform

Before integrating AI-powered features, configure the Antigravity AI Platform securely within the project.

Steps

- Step 1: Create an account on the Antigravity AI Platform.
- Step 2: Generate an API Key from the developer dashboard.
- Step 3: Store the API key securely inside the project's .env file.
ANTIGRAVITY_API_KEY=your_api_key_here
- Step 4: Load the environment variables using the dotenv package.
from dotenv import load_dotenv
load_dotenv()
- Step 5: Ensure the API key is never pushed to GitHub by adding .env to the .gitignore file.
.env

Activity 1.2: Research and Select AI Features

Before implementing AI functionality, analyze the project requirements and identify the AI capabilities needed.

The selected AI platform should be capable of:

- Analyzing users' financial information.
- Calculating financial health indicators.
- Predicting suitable debt settlement recommendations.
- Generating professional negotiation letters.
- Providing personalized financial recovery guidance.

The Antigravity AI Platform was selected because it supports intelligent financial analysis, natural language generation, and personalized recommendation systems suitable for debt management applications.

Activity 1.3: Define the System Architecture

Create the architecture diagram representing the interaction between all application components.

The architecture consists of:

- React.js frontend for user interaction.
- FastAPI backend for business logic and API processing.
- SQLite database using SQLAlchemy ORM.
- Antigravity AI Platform for intelligent recommendation generation.

The backend acts as the communication layer between the frontend, database, and AI platform, ensuring secure and efficient processing of financial information.

Activity 1.4: Set Up the Development Environment

Install and configure the required software before starting development.

Install Python

python --version

Create Virtual Environment

```
python -m venv venv
```

Activate Virtual Environment

Windows

```
venv\Scripts\activate
```

Install Backend Dependencies

```
pip install fastapi uvicorn sqlalchemy python-dotenv
```

Create React Project

```
npm create vite@latest frontend
```

Install React Dependencies

```
cd frontend
```

```
npm install
```

Install Additional Packages

```
npm install axios react-router-dom
```

Project Structure

```
frontend/  
  backend/  
    database/  
    models/  
    routes/  
    services/  
    utils/  
  .env  
  README.md
```


Milestone 2: Core Functionalities Development

Milestone 2 focuses on developing the core functionalities of the AI Powered Debt Relief & Financial Recovery Platform. In this phase, the application implements user authentication, loan management, financial analysis, AI-powered debt settlement recommendations, and negotiation letter generation. The FastAPI backend processes user requests, performs financial calculations, securely stores data in the SQLite database, and communicates with the Antigravity AI Platform to generate personalized financial insights.

Activity 2.1: Develop Core Financial Management Features

Implement the essential modules that provide users with complete debt management and financial recovery services.

User Authentication

- Develop secure user registration and login functionality.
- Validate user credentials before granting access.
- Implement password encryption and authentication mechanisms.
- Maintain user sessions securely.

Loan Management

- Allow users to add, update, view, and delete loan details.
- Store loan information including loan amount, EMI, interest rate, and due date.
- Display loan information on the user dashboard.

Financial Analysis

- Collect user financial information such as monthly income, monthly expenses, and outstanding debts.
- Calculate financial health indicators including debt-to-income ratio and repayment capability.
- Generate a financial health score for better decision-making.

AI Settlement Recommendation

- Send the user's financial information to the Antigravity AI Platform.
- Generate personalized debt settlement recommendations.
- Suggest suitable repayment strategies based on the user's financial condition.

Negotiation Letter Generation

- Generate professional loan settlement request letters using AI.
- Create personalized negotiation letters based on the user's financial profile.
- Allow users to review and download AI-generated letters.

Activity 2.2: Implement the FastAPI Backend

Develop the FastAPI backend to process requests, manage business logic, and integrate database and AI services.

Develop REST APIs

Implement REST APIs for the following functionalities:

- User Registration
- User Login
- Loan Management
- Financial Analysis
- AI Settlement Recommendation
- Negotiation Letter Generation

- History Management

Process User Requests

- Validate user input before processing.
- Handle API requests and responses efficiently.
- Perform financial calculations.
- Store user and loan information in the SQLite database.

Database Integration

- Connect the FastAPI backend with SQLite using SQLAlchemy ORM.
- Perform Create, Read, Update, and Delete (CRUD) operations.
- Maintain secure and efficient data storage.

AI Integration

- Send processed financial data to the Antigravity AI Platform.
- Receive AI-generated settlement recommendations and negotiation letters.
- Return AI responses to the frontend through REST APIs.

Error Handling

- Validate all user inputs.
- Handle invalid requests gracefully.
- Return meaningful error messages for API failures.
- Maintain application stability during AI communication failures.

Milestone 3: Backend API Development

Milestone 3 focuses on developing the backend application using **FastAPI**. The backend serves as the core of the **AI Powered Debt Relief & Financial Recovery Platform**, handling user authentication, loan management, financial analysis, AI integration, and database operations. REST APIs are developed to enable secure communication between the React frontend, SQLite database, and the Antigravity AI Platform.

Activity 3.1: Develop the Main Backend Application

Create the main FastAPI application that manages all API requests and integrates the different modules of the system.

Create the FastAPI Application

- Initialize the FastAPI project.
- Configure middleware and routing.
- Enable CORS to allow communication with the React frontend.
- Organize the project into modules such as routes, models, services, and database.

Develop Authentication APIs

Implement secure authentication endpoints for users.

- Create user registration API.
- Create user login API.
- Validate user credentials.
- Generate secure authentication tokens.
- Protect authorized routes from unauthorized access.

Develop Loan Management APIs

Create APIs to manage user loan information.

- Add new loan details.
- Retrieve loan information.
- Update existing loan records.
- Delete loan records.
- Validate financial data before storing it.

Develop Financial Analysis APIs

Implement APIs that analyze the user's financial information.

- Calculate monthly financial health.
- Compute debt-to-income ratio.
- Evaluate repayment capability.
- Generate financial health score.
- Return analysis results to the frontend.

Integrate Antigravity AI Platform

Connect the backend with the Antigravity AI Platform to provide intelligent financial assistance.

- Send financial information to the AI platform.
- Generate personalized debt settlement recommendations.
- Create AI-powered negotiation letters.
- Receive AI responses and return them through REST APIs.

Database Operations

Implement database operations using SQLite and SQLAlchemy ORM.

- Store user information.
- Save loan details.
- Maintain AI recommendation history.
- Store settlement records.
- Perform Create, Read, Update, and Delete (CRUD) operations efficiently.

API Validation and Error Handling

Ensure reliable communication by validating requests and handling errors.

- Validate all user inputs.
- Handle invalid API requests.
- Return appropriate HTTP status codes.
- Display meaningful error messages.
- Maintain secure and reliable API communication.

Testing the Backend APIs

Verify all backend services before integrating with the frontend.

- Test APIs using Postman.
- Verify database operations.
- Validate authentication.
- Test AI integration.
- Ensure accurate API responses.

This milestone completes the backend implementation of the application, enabling secure data management, financial analysis, AI-powered recommendations, and seamless communication between all system components.

Milestone 4: Frontend Development

Milestone 4 focuses on designing and developing a responsive, user-friendly interface for the **AI Powered Debt Relief & Financial Recovery Platform** using **React.js** and **Vite**. The frontend provides users with an intuitive experience for managing loan details, viewing financial analysis, and accessing AI-powered settlement recommendations. It communicates with the FastAPI backend through REST APIs and dynamically displays user data and AI-generated responses.

Activity 4.1: Designing and Developing the User Interface

Set Up the React Project

- Create the frontend application using React.js with Vite.
- Organize the project into reusable components and pages.
- Configure React Router for navigation between different pages.

Develop Authentication Pages

- Design the User Registration page.
- Develop the Login page with input validation.
- Implement secure authentication flow between frontend and backend.

Create Dashboard Interface

- Design a responsive dashboard displaying financial information.
- Display loan details, financial health score, and AI recommendations.
- Provide easy navigation to all application modules.

Develop Loan Management Pages

- Create forms for adding and updating loan information.
- Display existing loan records in a structured format.
- Allow users to edit or delete loan details.

Design Financial Analysis Page

- Display financial metrics such as monthly income, expenses, and debt-to-income ratio.
- Present AI-generated settlement recommendations in an easy-to-understand format.
- Show negotiation letter generation options.

Responsive UI Design

- Use CSS and React components to create a responsive layout.
- Ensure compatibility across desktop, tablet, and mobile devices.
- Maintain a clean and user-friendly interface.

Activity 4.2: Integrating Frontend with Backend APIs

Connect REST APIs

- Use Axios to communicate with FastAPI backend services.
- Fetch user information, loan details, and financial analysis data.
- Display API responses dynamically on the frontend.

Handle User Input

- Validate all form inputs before submission.
- Display success and error messages based on API responses.
- Prevent invalid data from being submitted.

Display AI-Generated Results

- Show AI-powered settlement recommendations.

- Display financial health analysis.
- Present AI-generated negotiation letters in a readable format.

Manage Application State

- Maintain user session after successful login.
- Update dashboard information dynamically.
- Refresh displayed data after loan updates or AI analysis.

Testing the User Interface

- Verify navigation between pages.
- Test responsiveness on different screen sizes.
- Ensure smooth communication between frontend and backend.
- Validate that all user actions produce the expected results.

Milestone 5: Deployment

Milestone 5 focuses on deploying the AI Powered Debt Relief & Financial Recovery Platform after successful development and testing. This phase involves configuring the production environment, installing all required dependencies, deploying both the frontend and backend applications, and verifying that the application functions correctly in a live environment. Proper deployment ensures that users can securely access the platform from any location through a web browser.

Activity 5.1: Preparing the Application for Deployment

Before deployment, configure the application and verify that all project dependencies are installed correctly.

Create and Activate a Virtual Environment

Create a virtual environment for the backend project.

```
python -m venv venv
```

Activate the virtual environment.

Windows

```
venv\Scripts\activate
```

Install Backend Dependencies

Install all required Python packages.

```
pip install -r requirements.txt
```

If the requirements file is unavailable, install the necessary packages manually.

```
pip install fastapi uvicorn sqlalchemy python-dotenv
```

Install Frontend Dependencies

Navigate to the frontend directory and install the required packages.

```
cd frontend
```

```
npm install
```

Configure Environment Variables

Create a .env file in the backend project directory and add the required API credentials.

```
ANTIGRAVITY_API_KEY=your_api_key_here
```

```
DATABASE_URL=sqlite:///debt_platform.db
```

Ensure that the .env file is included in the .gitignore file to protect sensitive information.

Activity 5.2: Local Testing and Deployment

Start both the backend and frontend servers to verify that the application works correctly.

Run the FastAPI Backend

```
uvicorn main:app --reload
```

The backend will be available at:

```
http://127.0.0.1:8000
```

Run the React Frontend

```
npm run dev
```

The frontend will be available at:

<http://localhost:5173>

Verify Application Functionality

Test all major features of the application before deployment.

- Verify user registration and login.
- Test loan management functionalities.
- Validate financial analysis calculations.
- Check AI-generated settlement recommendations.
- Verify negotiation letter generation.
- Ensure all frontend pages communicate successfully with the backend APIs.

Activity 5.3: Cloud Deployment

Deploy the application to a cloud hosting platform for public access.

Backend Deployment

- Deploy the FastAPI backend on a cloud platform such as Render or Railway.
- Configure environment variables and API credentials.
- Verify backend API accessibility.

Frontend Deployment

- Deploy the React application using Vercel or Netlify.
- Connect the frontend with the deployed backend APIs.
- Verify all application features after deployment.

Final Verification

- Test the complete application using the deployed URL.
- Verify database connectivity.
- Ensure AI-generated recommendations and negotiation letters function correctly.
- Confirm that the application performs efficiently across different devices and browsers.

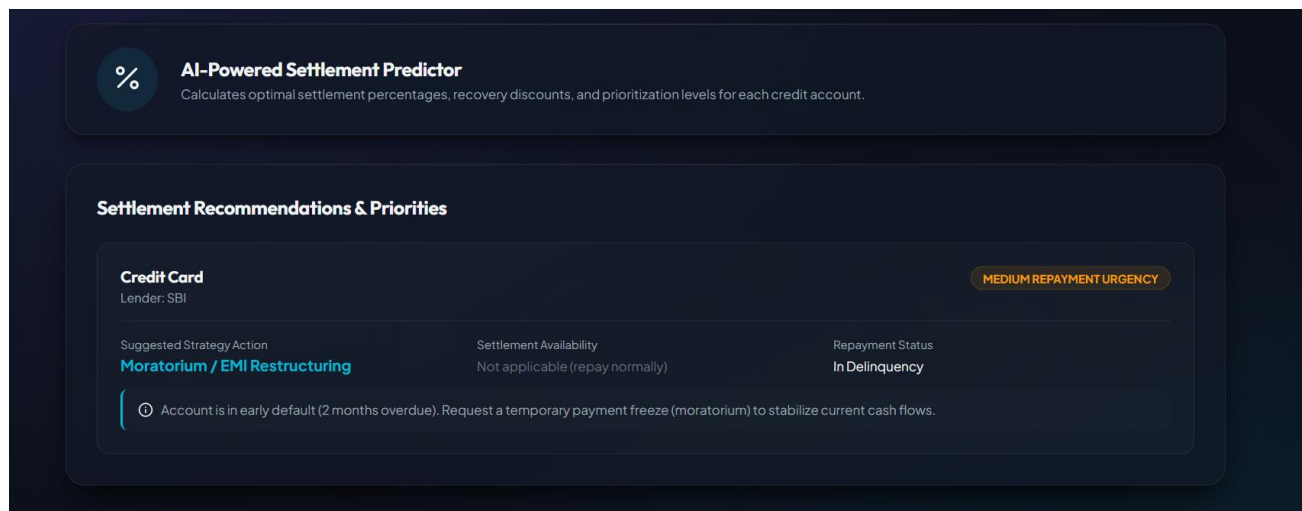
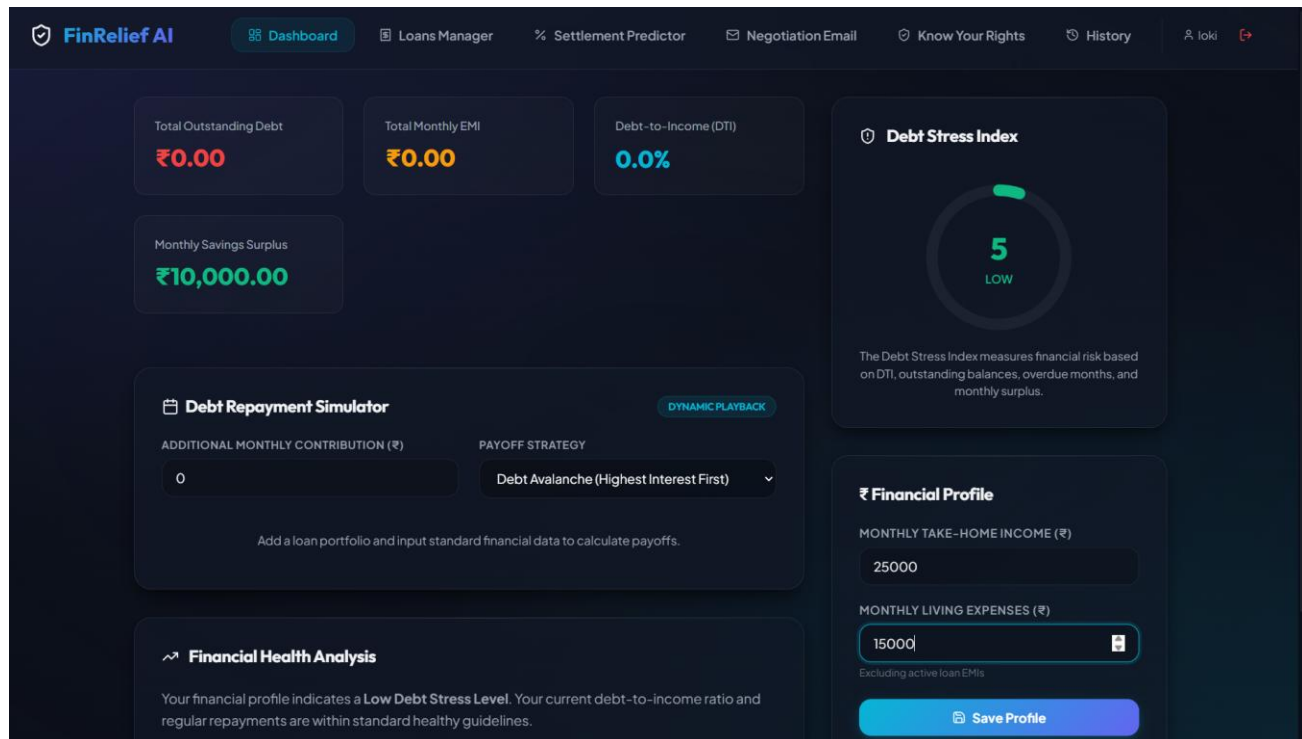
Important Notes:

- Ensure that the ANTIGRAVITY_API_KEY and other environment variables are correctly configured in the .env file before running the application.
- Never upload the .env file or API keys to a public GitHub repository. Add .env to the .gitignore file for security.
- Verify that the React frontend, FastAPI backend, SQLite database, and Antigravity AI Platform are properly connected before deployment.
- Thoroughly test all major functionalities, including user authentication, loan management, financial analysis, AI recommendations, and negotiation letter generation.
- After deployment, monitor the application regularly to ensure stable performance and resolve any errors or API issues promptly.

Exploring the Website's Web Pages:

Home / Generator Page:

Description:



 **Settlement Action Plan**

Credit Card (SBI)
Account is in early default (2 months overdue). Request a temporary payment freeze (moratorium) to stabilize current cash flows.

MORATORIUM / EMI RESTRUCTURING

 **AI Negotiation Email Generator**

SELECT LOAN ACCOUNT

-- Choose Account --

NEGOTIATION STRATEGY

☒ Financial Hardship Concession

☐ One-time Lump-sum Settlement

☐ Interest Rate Reduction

☐ Tenure Extension Request

Generate Letter

Input Section:

Description:

Results Section:

Description:

+ Add Loan Details

LOAN TYPE / NAME

Credit Card

LENDER NAME

SBI

OUTSTANDING BALANCE (₹)

12000

MONTHLY EMI (₹)

2000

INTEREST RATE (%)

15

OVERDUE DURATION (MONTHS)

2

+ Add Loan Account

Conclusion:

The **AI Powered Debt Relief & Financial Recovery Platform** is an intelligent web application developed using **React.js, FastAPI, SQLite, SQLAlchemy, and the Antigravity AI Platform** to assist users in managing debt and improving their financial well-being. The platform provides secure user authentication, efficient loan management, financial analysis, AI-powered settlement recommendations, and automated negotiation letter generation through an intuitive and responsive interface.

Throughout the project, key phases such as system architecture design, backend and frontend development, database integration, AI implementation, testing, and deployment were successfully completed. The integration of the **Antigravity AI Platform** enhances the application's ability to deliver personalized financial insights and settlement strategies, helping users make informed financial decisions.

In the future, the platform can be further enhanced by incorporating support for multiple financial institutions, real-time credit score analysis, multilingual support, advanced analytics dashboards, and stronger security features. These enhancements would improve scalability, user experience, and overall effectiveness, making the platform a comprehensive solution for AI-driven debt management and financial recovery.